

Santiago Martínez Novoa

Systems and Computing Engineer

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EDUCATION

Universidad de los Andes <i>M.Sc. Information Engineering</i>	Bogotá, Colombia 2025 – 2026 (To be conferred)
Universidad de los Andes <i>B.Sc. Systems and Computing Engineering (Cum Laude), Minor in French</i>	Bogotá, Colombia 2021 – 2024

EXPERIENCE

Mitacs Globalink Research Intern - Collaborating Researcher <i>Ontario Tech University</i>	2026-05 – Present Oshawa, Canada
- Co-authored a Findings of EMNLP 2026 paper on adaptive language sampling; leading its follow-up on when to stop cross-lingual realignment, under review at ARR - Co-developing M2-ALIGN: extending vision-language models to 11 languages while preserving their multilingual reasoning	
Data Scientist <i>EdTech Node, Universidad de los Andes</i>	2026-03 – Present Bogotá, Colombia
- Lead the unit's adoption of data governance and quality policies, enabling university-wide data products such as academic percentiles and dropout indicators - Lead the team's research line: dropout-prediction ML models behind 2 peer-reviewed publications (AIED & WEEF 2026)	
Graduate Teaching Assistant <i>Universidad de los Andes</i>	2025-06 – Present Bogotá, Colombia
- Designed the hands-on curriculum for two sequential master's courses on RAG systems and AI agents, reaching 200+ students across 3 semesters - Built tutorials, projects, and RAGAS-based evaluation pipelines with LangChain, LangGraph, and LangSmith	
Data Scientist - Graduate Research Assistant <i>Vice Dean's Office of Student Affairs, Universidad de los Andes</i>	2025-01 – 2026-03 Bogotá, Colombia
- Implemented complete medallion architecture in Databricks, fully automating institutional data pipelines - Optimized ETL pipeline performance by 87% by migrating from Pandas to Polars, reducing processing time from 15 to 2 minutes - Replaced external counseling platform, saving the university ~\$100K USD annually - Co-implemented system alerting over 1000 critical student cases	
Generative Artificial Intelligence for Research - Research Assistant <i>Universidad de los Andes</i>	2025-01 – 2025-06 Bogotá, Colombia
- Built locally-deployed RAG systems using Ollama and Gemini API for domain-specific research questions - Designed LLM evaluation architecture using Prometheus, BERT score and OpenAI API for response quality assessment - Developed custom AI agents to optimize university research processes - Automated student assessment using LLMs aligned with rubrics and expected answers	
Machine Learning Intern <i>Precia PPV S.A</i>	2024-08 – 2024-12 Bogotá, Colombia
- Developed MLP model reducing financial calculation time by 70% for client applications - Implemented end-to-end AWS infrastructure with automated ETL pipelines using Glue and Lambda - Presented our work on models and AWS infrastructure at Analytics Forum 2025	
Summer Research Intern <i>College of Computing and Information Science, Cornell University</i>	2024-06 – 2024-08 Ithaca, United States

- Designed Data Discovery architecture to explore and modify datasets in various formats
- Developed upload and scoring system through semantic search across multiple databases

TECHNICAL SKILLS

- *Programming Languages:* Python, SQL, JavaScript, TypeScript
- *Machine Learning & AI:* Scikit-learn, Keras, PyTorch, TensorFlow
- *LLM & Natural Language Processing:* Transformers, BERT, LangChain, LangGraph, LangSmith, Ollama
- *Data Processing:* Pandas, Polars, Databricks
- *Cloud & DevOps:* AWS, Azure Functions, Azure Blob Storage
- *Databases:* PostgreSQL, MongoDB, ChromaDB, Pinecone